CALCULUS · UNIT 7: APPLICATIONS OF INTEGRALS

Glass Vase Studio

Use area between curves and volumes of revolution to design glass vases and bottles.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Lumen Glass Studio makes vases, bottles, and paperweights. Each design is a curve spun around an axis, the logo is a region between two curves, and glass is priced by volume.

You will use integrals to calculate the area and volume of each design and price the products.



Glass Price

$0.08 per cm³

Vase Profile

$r = 4 + \sin(\pi x/10)$

x in cm, $0 \le x \le 20$

Learning Goals

- I can find the area between two curves.
- I can find a volume with the disc method.
- I can use an integral to find a total from a rate.

Project Timeline

When	What you will do
Day 1	Studio Logo
Day 2	Bud Vase
Day 3	Production
Final	Product Design Portfolio + reflection

Part 1: Studio Logo

Day 1 · The logo is the region between y = √x and y = $\frac{x}{2}$.

Design Specification Cards (units: cm)

Logo	Bud Vase	Tall Vase
Between y = √x	y = √x rotated about	r = 4 + sin(πx/10)
and y = x/2	the x-axis, 0 ≤ x ≤ 16	0 ≤ x ≤ 20

Hollow Vase	Paperweight	Production
Outer r = 4 + sin(πx/10)	Base: y = 4 − x², y ≥ 0	Rate p(t) = 6 + 2t
inner r = 3.5 + sin(πx/10)	square cross sections	vases per hour

1. Find where the curves intersect.

Starter: *Set √x = $\frac{x}{2}$ and square both sides.*

2. Which curve is on top between x = 0 and x = 4?

3. Find the area of the logo.

4. The logo is etched at $1.50 per cm². Find the cost.

Part 2: Bud Vase

Day 2 · Rotate $y = \sqrt{x}$, $0 \le x \le 16$, about the x-axis.

5. Write the volume integral using the disc method.

Starter: $V = \pi \int [r(x)]^2 \, dx$

6. Evaluate the volume.

7. Find the cost of the glass at \$0.08 per cm^3 (solid paperweight version).

8. Why is $r(x) = \sqrt{x}$ the radius?

Part 3: Production

Day 3 · The studio makes $p(t) = 6 + 2t$ vases per hour for $0 \le t \le 4$.

9. Find the total number of vases made in 4 hours.

10. Find the average production rate.

11. Design a logo with two curves of your own and find its area.

Final Product: Product Design Portfolio

Your final product must include:

- Logo area and cost
- Bud vase volume and cost
- Production total and average
- Your own logo
- Design portfolio

Reflection

R1. Why do you subtract the bottom curve from the top curve?

R2. Why does the disc method square the radius?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Areas and volumes are set up and evaluated correctly and used to price and design products. Goes beyond the minimum with a creative, realistic choice.	Areas and volumes are set up and evaluated correctly and used to price and design products.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Portfolio includes diagrams, integrals, results, and prices. Polished, organized, and visually engaging.	Portfolio includes diagrams, integrals, results, and prices. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.